

IN AMENAS, Algeria

WMO: 606110

Lat: 28.052N

Lon: 9.643E

Elev: 563

StdP: 94.74

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	2.8	-12.9	1.3	13.7	-11.1	1.6	14.4	14.7	13.2	12.7	13.9	2.5	60	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	43.1	19.7	42.0	19.6	40.9	19.4	23.6	37.1	22.4	36.5	21.4	35.9	4.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.9	31.6	17.5	13.4	30.5	16.1	12.2	29.8	73.4	36.7	68.6	36.2	64.5	35.8	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
13.2	11.2	9.9	DB	-1.5	44.8	1.6	0.8	-2.6	45.4	-3.6	45.9	-4.5	46.3	-5.7	46.9	
			WB	-3.7	24.3	1.2	2.8	-4.5	26.3	-5.2	27.9	-5.9	29.5	-6.8	31.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.0	11.0	13.3	18.1	23.5	28.5	31.8	32.7	32.4	30.3	24.4	17.3	12.3	
	DBStd	8.51	2.89	3.54	4.10	4.06	3.32	2.59	2.13	1.85	2.43	3.24	3.28	3.00	
	HDD10.0	41	21	8	1	0	0	0	0	0	0	0	1	10	
	HDD18.3	681	228	146	55	6	1	0	0	0	0	1	57	187	
	CDD10.0	4791	51	100	252	405	573	654	703	694	610	446	221	82	
	CDD18.3	2391	1	5	48	161	315	404	445	436	360	189	26	2	
	CDH23.3	36853	17	118	715	2183	4603	6553	7483	7157	5396	2226	363	38	
CDH26.7	23366	2	27	272	1133	2825	4426	5189	4863	3431	1098	94	6		
Wind	WSAvg	4.5	3.8	4.3	4.7	5.1	5.5	5.1	4.8	4.6	4.4	4.3	3.7	3.8	
Precipitation	PrecAvg	20	2	1	4	2	2	1	0	1	1	3	1	2	
	PrecMax	75	17	8	35	20	26	30	0	14	15	29	9	12	
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0	
	PrecStd	17	4	2	8	4	4	5	0	2	3	7	2	3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	29.9	34.8	39.2	41.9	44.1	44.5	43.8	41.8	37.8	31.3	27.1	
		MCWB	11.7	14.0	15.6	17.7	18.9	19.0	20.3	20.0	19.6	18.5	15.3	12.6	
	2%	DB	22.7	26.8	32.1	36.9	40.0	42.7	43.2	42.1	40.8	35.8	29.1	23.9	
		MCWB	10.7	12.4	15.0	16.3	18.5	19.0	20.0	20.1	19.2	18.3	14.7	12.0	
	5%	DB	20.4	24.1	29.8	34.9	38.5	41.2	42.1	41.0	39.6	34.1	27.2	21.4	
		MCWB	9.7	11.3	14.2	16.0	18.1	19.0	19.8	19.9	19.1	17.7	14.1	11.0	
	10%	DB	18.3	21.8	27.4	32.9	36.9	39.9	40.8	39.9	38.2	32.4	25.2	19.2	
		MCWB	9.0	10.4	13.3	15.4	17.7	18.9	19.6	19.6	19.0	17.4	13.5	10.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	15.3	18.6	20.3	23.2	24.6	24.2	25.2	24.1	22.4	18.8	14.8
			MCDB	21.9	24.3	29.8	33.4	35.8	37.2	39.6	37.7	35.5	33.0	27.8	21.4
2%		WB	11.6	13.5	16.6	18.5	21.0	22.6	22.8	23.3	22.5	20.4	16.4	13.0	
		MCDB	20.5	23.5	27.5	31.5	34.6	35.9	39.1	37.2	34.9	31.4	25.8	21.8	
5%		WB	10.3	12.1	15.1	17.3	19.5	21.1	21.4	22.0	21.2	19.1	15.0	11.8	
		MCDB	18.6	22.1	26.9	30.7	33.5	35.3	37.9	36.7	34.4	30.4	24.9	20.0	
10%		WB	9.4	10.9	13.8	16.2	18.5	19.9	20.3	20.8	20.1	18.1	13.9	10.7	
		MCDB	17.5	20.6	26.2	29.8	34.1	35.6	37.5	36.7	33.4	29.5	23.8	18.5	
Mean Daily Temperature Range	5% DB	MDBR	14.3	14.8	15.3	15.0	14.6	14.6	15.1	14.6	14.4	14.0	14.5	13.8	
		MCDBR	16.7	18.1	17.9	17.2	16.2	15.9	16.2	15.5	15.8	15.7	16.4	16.2	
		MCWBR	9.0	9.4	8.8	7.6	6.7	6.2	6.7	6.2	6.3	6.8	8.0	8.6	
	5% WB	MCDBR	14.9	15.8	15.8	14.7	14.1	14.1	14.9	14.2	13.7	13.6	14.5	14.5	
		MCWBR	8.8	9.1	8.9	7.5	7.2	7.4	7.6	7.3	6.6	6.9	7.9	8.5	
Clear-Sky Solar Irradiance	taub	0.353	0.385	0.488	0.555	0.610	0.585	0.510	0.504	0.542	0.495	0.405	0.361		
	taud	2.150	2.047	1.776	1.652	1.575	1.638	1.805	1.831	1.748	1.834	2.023	2.140		
	Ebn at Noon	874	875	802	762	722	736	792	794	746	751	804	843		
	Edh at Noon	129	156	218	254	275	257	217	209	221	191	146	125		
All-Sky Solar Radiation	RadAvg	4.28	5.13	6.18	6.93	7.36	7.79	8.05	7.49	6.49	5.30	4.37	3.89		
	RadStd	0.20	0.21	0.29	0.29	0.30	0.26	0.11	0.18	0.23	0.28	0.14	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.89	+0.45	-1.66	-1.93	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	-1.36	N/A	-1.13	-1.53	N/A	N/A	N/A	N/A

Nomenclature: See separate page